AY25/26 BIO220 Coursework1: Structure Prediction and Comparative Analysis of the Human μ-Opioid Receptor

Total Marks: 100 (15% of the Module)

Protein of Interest: Human μ-opioid receptor

The **μ-opioid receptor (MOR)** is a G-protein-coupled receptor (GPCR) that plays a critical role in pain regulation and is the primary target of opioid analgesics such as morphine. Understanding its structure and interactions with ligands is essential for drug discovery and the development of safer analgesics.

Instructions

In this coursework, you will use the skills learned in the tutorial to build a comprehensive structural analysis of the human MOR. You will progress through three stages:

1. **Structure Prediction:** Predicting the 3D structure of MOR from its amino acid sequence using two methods.
2. **Comparative Analysis:** Performing a structural comparison between your predicted models and an experimentally determined structure.
3. **Critical Discussion:** Evaluating the performance of the different models and discussing the biological implications of your findings.

Marking scheme

Marks will be awarded based on the following criteria:

1. Accuracy and completeness of answers.
2. Producing high-quality molecular visualization figures.
3. Clear and concise captions for the figures.
4. Correct submission of all required files.

Question 1: Structure Prediction and Evaluation [40 marks]

1.1 (10 marks) Retrieve the full-length amino acid sequence for the human μ-opioid receptor from UniProt (Accession ID: P35372). Using this sequence, generate a 3D structural model using both of the following web servers:

- **AlphaFold** (You may use the pre-computed model in the AlphaFold Database).
- **SWISS-MODEL**.

1.2 (15 marks) For both your AlphaFold and SWISS-MODEL structures, generate a figure showing the protein colored by the model's confidence score (pLDDT for AlphaFold, QMEAN for SWISS-MODEL).

- **Include both figures.**
- In the figure captions, briefly **explain what the confidence score represents** for each method.
- Identify the regions with the **lowest confidence** in both models. Based on the known topology of a GPCR, provide a biochemical reason why these specific regions (e.g., N/C-termini, intracellular/extracellular loops) are often more flexible and thus harder to predict accurately.

1.3 (15 marks) Download the experimentally determined structure of MOR from the PDB (PDB ID: 8EF6). This structure represents the receptor in an active state.

- Using VMD or another molecular viewer, create a high-quality figure of the 8EF6 biological assembly.
- In your figure, **visually distinguish the seven transmembrane helices** from the loops using different colors or representations.
- The PDB entry contains more than just the receptor. Identify the co-complexed molecules (e.g., the ligand, the G-protein/nanobody). **Label these components** in your figure.
- In your figure caption, briefly **explain why an experimentalist would include these additional molecules** when trying to solve the structure of an *active* receptor.

Question 2: Structural Comparison [40 marks]

2.1 (15 marks) Perform a structural alignment of your two predicted models (AlphaFold, SWISS-MODEL) against the experimental structure (PDB: 8EF6) using the RCSB PDB Alignment Tool or similar software.

- Create a table reporting the **RMSD** and **TM-score** for the alignment of each model against the experimental structure.
- Based on these quantitative metrics, which model is a better fit for the overall fold of the experimental structure?
- Generate a single, high-quality figure showing all three structures superimposed. Use different colors for each structure for clarity.

2.2 (15 marks) Even without docking, we can assess how well the models replicate the site of function. Focus on the orthosteric ligand-binding pocket of the MOR.

- Create a figure that shows a close-up view of the binding pocket in the **experimental structure (8EF6)** with its bound ligand. Identify and label **at least two key residues** that interact with the ligand.
- Create a second figure comparing a close-up of the binding pockets from your **AlphaFold and SWISS-MODEL structures**.
- Comment on the similarities and differences. Do the residues in the predicted models have similar backbone conformation? Are the key residues you identified oriented in a similar way as in the experimental structure?

2.3 (10 marks) The intracellular loops (ICLs) of a GPCR are critical for binding to G-proteins and initiating signaling.

- Compare the conformation of the intracellular loops (especially ICL3) between the experimental active-state structure and your two predicted models.
- Describe any significant differences you observe. Are the loops in your models structured similarly to the active state, or are they in a different conformation?

Question 3: Critical Discussion [20 marks]

3.1 (10 marks) Write a summary (200-250 words) critically evaluating the success of the structure prediction methods for this specific target. Your discussion should address:

- Which prediction method (AlphaFold or SWISS-MODEL) produced a more functionally relevant model of the MOR?
- Why do you think one model performed better than the other? (Consider the fundamental differences in how AlphaFold and homology modeling work).

3.2 (10 marks) Write a conclusion about biological states for this specific target. The experimental structure 8EF6 is in an 'active' state. Based on your structural comparisons above, do your predicted models appear to represent an active or an inactive state?

- Justify your conclusion by synthesizing **two distinct pieces of structural evidence** from your analysis. One piece of evidence should relate to the conformation of the **intracellular face** (from Question 2.3), and the other should be a major conformational change in the **transmembrane core** (e.g., the outward movement of TM6) characteristic of GPCR activation.

Final Submission Checklist

You must submit a single report in PDF format containing all your answers, figures, and captions. Additionally, you must upload the following files in a zipped folder:

1. [your_name]_AlphaFold.pdb: The structure file from AlphaFold.
2. [your_name]_SwissModel.pdb: The structure file from SWISS-MODEL.
3. [your_name]_Alignment_Session.vmd (or .pse for PyMOL): A visualization session file showing your final superimposed structures, allowing the marker to replicate your view from Question 2.1.

Failure to submit these files will result in a penalty.